assignment4 - Gauss's law - with tutorials!

This is a preview of the published version of the quiz

Started: Mar 15 at 3:11pm

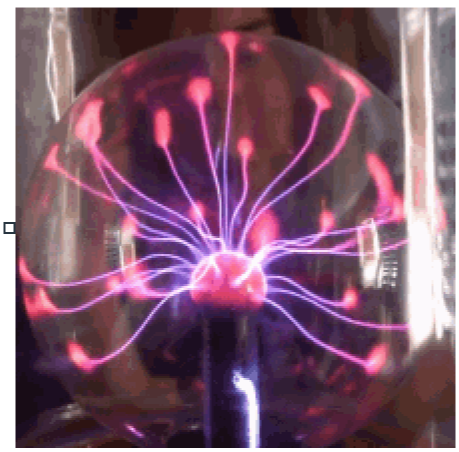
Quiz Instructions

Its absolutely crucial not to look at the solutions.

There are tutorials in dropbox. Do not hesitate to pause the video to see if you can get the right answers. Work with a study group.

hint: if you are asked to find the flux. Then use flux = Qin/ε_0 .

if you are asked to find the electric field E, then use sum(E x area) = Qin/ε_0



Question 1 8.4 pts

Two charges of 15 pC and –40 pC are inside a cube with sides that are of 0.40-m length. Determine the net electric flux through the surface of the cube.
hint: there is a video tutorial in the shared folder.

- -1.1 N · m^2/C
- +1.1 N · m^2/C
- –2.8 N · m^2/C
- 2.8 N · m^2/C

Question 2 8.4 pts

A uniform linear charge density of 4.0 nC/m is distributed along the entire *x* axis. Consider a spherical (radius = 5.0 cm) surface centered on the origin. Determine the electric flux through this surface.
hint: there is a video tutorial

- ○ 45 N · m^2/C
- ○ 62 N · m^2/C
- ○ 79 N · m^2/C
- ○ 23 N · m^2/C

Question 3 8.4 pts

The *xy* plane is "painted" with a uniform surface charge density which is equal to 40 nC/m^2. Consider a spherical surface with a 4.0-cm radius that has a point in the *xy* plane as its center. What is the electric flux through that part of the spherical surface for which $z > 0$?
hint: there is a video tutorial

- ○ 20 N · m^2/C
- ○ 23 N · m^2/C
- ○ 11 N · m^2/C
- ○ 17 N · m^2/C

Question 4 8.4 pts

A long cylinder (radius = 3.0 cm) is filled with a nonconducting material which carries a uniform charge density of 1.3 μC/m^3. Determine the electric flux through a spherical surface (radius = 2.0 cm) which has a point on the axis of the cylinder as its center.
hint: there is a video tutorial

- ○ 4.9 N · m^2/C
- ○ 15 N · m^2/C
- ○ 6.4 N · m^2/C

○

7.2 N · m^2/C

Question 5 8.4 pts

Two infinite parallel surfaces carry uniform charge densities of 0.20 nC/m^2 and –0.60 nC/m^2. What is the magnitude of the electric field at a point between the two surfaces?
hint: there is a video tutorial

○

34 N/C

○

17N/C

○

23N/C

○

45 N/C

Question 6 8.4 pts

A long nonconducting cylinder (radius = 12 cm) has a charge of uniform density (5.0 nC/m^3) distributed throughout its column. Determine the magnitude of the electric field 5.0 cm from the axis of the cylinder.
hint: tutorial video

○

14 N/C

○

34 N/C

○

31 N/C

○

20N/C

Question 7 8.4 pts

Charge of uniform density (80 nC/m^3) is distributed throughout a hollow cylindrical region formed by two coaxial cylindrical surfaces of radii 1.0 mm and 3.0 mm. Determine the magnitude of the electric field at a point which is 4.0 mm from the symmetry axis.
hint: tutorial video

○

17 N/C

○

8 N/C

○

10 N/C

○
9.0 N/C

Question 8 8.4 pts

A solid nonconducting sphere (radius = 12 cm) has a charge of uniform density (30 nC/m^3) distributed throughout its volume. Determine the magnitude of the electric field 15 cm from the center of the sphere.

hint: draw the situation and notice that 15cm > 12cm/ So you have a spherical bag (sphere of area 4piR^2 also called Gaussian surface) with a solid sphere inside. The solid sphere has a smaller radius with charge inside (its not a conductor)/
Qin = density x (volume of the sold sphere inside). Volume is 4/3 pi r^2
so apply Gauss law. The gaussian surface is the big sphere (4 pi R^2 = A)
so E x A = Qin/E0 with Qin = density x volume small solid sphere.

○
87 N/C

○
22N/C

○
49N/C

○
31 N/C

Question 9 8.4 pts

A point charge of 6.0 nC is placed at the center of a hollow spherical conductor (inner radius = 1.0 cm, outer radius = 2.0 cm) which has a net charge of –4.0 nC. Determine the resulting charge density on the inner surface of the conducting sphere.
Next question 9B would be find the charge on the outside surface. I skip but try. I cover in the video

hint: video tutorial.

○
–9.5 μC/m^2

○
+9.5 μC/m^2

○
–8.0 μC/m^2

○
–4.8 μC/m^2

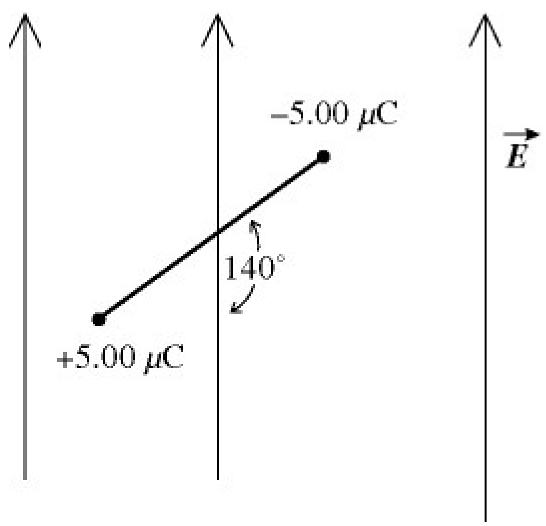
Question 10 8.4 pts

An astronaut is in an all-metal chamber outside the space station when a solar storm results in the deposit of a large positive charge on the station. Which statement is correct?

- The astronaut must abandon the chamber if the electric field on the outside surface becomes greater than the breakdown field of air.
- The astronaut must abandon the chamber immediately to avoid being electrocuted.
- The astronaut must abandon the chamber immediately because the electric field inside the chamber is non-uniform
- The astronaut does not need to worry: the charge will remain on the outside surface.

Question 11 8.4 pts

An electric dipole consists of charges ±5.00 μC separated by 1.20 mm. It is placed in a vertical electric field of magnitude 525 N/C oriented as shown in the figure. The magnitude of the net torque this field exerts on the dipole is closest to



- 1.21×10^{-6} N • m.
- 2.02×10^{-6} N • m.
- 2.41×10^{-6} N • m.
- 1.01×10^{-6} N • m.
- 3.15×10^{-6} N • m.

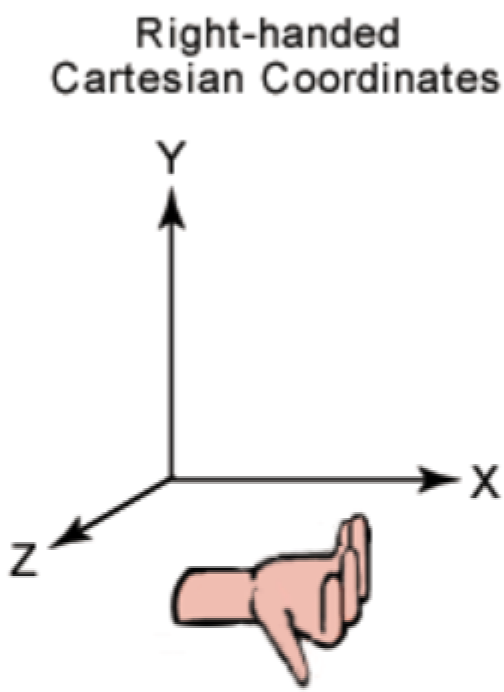
Question 12 8.4 pts

An initially-stationary electric dipole of dipole moment p = (5.00×10^{-10} C • m) **i** (so along the x-axis) placed in an electric field E = (2.00×10^{6} N/C) i + (2.00×10^{6} N/C) j. so E (2 10^6, 210^6) What is the magnitude of the maximum torque that the electric field exerts on the dipole?

hint: you can use the definition with the magnitudes of the troque |p x E |= |p| |E| sin(angle)

Trace E in a (x,y) coordinate. Notice that x = y for E. So E makes a 45 degrees with the x-axis. p is along the x-axis. So the angle between E and p is therefore ...... (trace). Then it is easy. The torque is along the z-axis. (use right hand coordinate system).

An alternative is to use Calculus 3 skils . i x j = k and i x i = 0.



○ 0.00 N · m

○ 1.00×10^{-3} N • m

○ 1.40×10^{-3} N • m

- ○ 2.80×10^{-3} N • m
- ○ 2.00×10^{-3} N • m

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